

# Sparse convex hulls for network flows coupled to a simplex

## Abstract

We study the convex hull of selected bilinear products between a bounded equality-constrained network flow and a simplex vector. The network and the retained product coordinates provide two distinct sources of structure: circulations factor across undirected blocks, and each block need retain only the simplex states observed there. We develop exact formulations and original-coordinate descriptions from these reductions, and examine the limits of small-coefficient descriptions. The starting point is classical simplex-vertex disaggregation; the structural question is how much of that formulation is needed for a sparse set of products.

## 1 Introduction

Network models often combine continuous routing decisions with a choice among mutually exclusive services or operating modes. A route-specific adjustment to the cost of a service gives a product of an arc flow and a service weight. Only a small fraction of the possible route-service pairs may occur in the model. Convexifying each product independently discards restrictions linking flows through the network and linking service weights through the simplex. An exact joint hull retains these restrictions, but its standard extended formulation introduces a copy of every arc flow for every simplex vertex.

This paper studies the structure of that joint hull when only selected products are retained. We distinguish the size of an extended formulation from the coefficients and separation cost of an original-coordinate description. We also distinguish equality-constrained flow models from network models with balance inequalities: adding slack arcs to convert the latter to the former can change the graph structure on which a specialized result depends. The elementary foundations below make the state and graph decompositions explicit, including degenerate capacities and zero simplex weights.

## 2 Model, prior work, and disaggregation

### 2.1 The sparse network-simplex hull

Let  $D = (V, E)$  be a finite directed multigraph. Parallel arcs and self-loops are permitted. Its incidence matrix  $A \in \{-1, 0, 1\}^{V \times E}$  has a  $-1$  at the tail and a  $+1$  at the head of a non-loop arc; a self-loop has a zero column. Thus  $Ax = b$  uses the convention *incoming minus outgoing*. For  $b \in \mathbb{Q}^V$  and finite  $u \in \mathbb{Q}_+^E$ , set

$$P = \{x \in \mathbb{R}^E : Ax = b, 0 \leq x \leq u\}. \quad (1)$$

Lower bounds can be handled by translating  $x$  and the corresponding product coordinates. Throughout, graph assumptions concern the graph in (1) after any such modeling transformations.

For an integer  $m \geq 0$ , write  $[m] = \{1, \dots, m\}$  and

$$\Delta_m = \{y \in \mathbb{R}_+^m : \mathbf{1}^\top y \leq 1\}, \quad \lambda_j = y_j \ (j \in [m]), \quad \lambda_0 = 1 - \mathbf{1}^\top y. \quad (2)$$

The *states* are the  $d = m + 1$  vertices  $e^0 = 0, e^1, \dots, e^m$  of  $\Delta_m$ ;  $e^j$  is the  $j$ th unit vector when  $j \geq 1$ . The residual state is state 0. An observation set  $\mathcal{O} \subseteq E \times [m]$  specifies the products retained as original coordinates. We study

$$S = \{(x, y, z) : x \in P, y \in \Delta_m, z_{ej} = x_e y_j \quad ((e, j) \in \mathcal{O})\}, \quad (3)$$

$$H = \text{conv } S. \quad (4)$$

An unobserved product is absent from  $z$ , even if it is introduced temporarily in an extended formulation. State 0 has no observed coordinates. We use  $\text{conv } \emptyset = \emptyset$ . When  $m = 0$ ,  $y$  and  $z$  are empty vectors and  $H = P$ .

The original coordinates are  $(x, y, z)$ . An extended formulation is a linear system in these coordinates and additional continuous variables whose projection is  $H$ . A separation procedure must either certify membership or produce a linear inequality valid for all of  $H$  and strictly violated by the candidate. Unless specified otherwise, a facet is taken relative to the affine hull. A bound on integer coefficients presupposes a stated normalization; a ratio of two nonzero coefficients is invariant under scaling the inequality. Arithmetic-operation bounds are distinguished from bounds on rational encoding length.

For comparison, the usual independent product relaxation adds, for every  $(e, j) \in \mathcal{O}$ , the four inequalities

$$0 \leq z_{ej}, \quad z_{ej} \leq u_e y_j, \quad z_{ej} \leq x_e, \quad z_{ej} \geq x_e - u_e(1 - y_j) \quad (5)$$

to  $x \in P$  and  $y \in \Delta_m$ . These are the McCormick inequalities for the box  $[0, u_e] \times [0, 1]$  [14]. They need not describe the joint hull in (4).

## 2.2 Relation to prior work

The general convexification principle is established. Disjunctive hull formulations [3] apply after expanding the simplex into its vertices. Simultaneous convexification over a polytope times a simplex is developed by Davarnia et al. [5]. Khademnia and Davarnia [11] specialize this setting to network polytopes: their electronic companion, p. 1, equation (25), gives a full extended hull,<sup>1</sup> and their aggregation-and-projection framework gives a complete original-space description in principle. Their explicit tree and forest constructions identify more structured inequality families; the forest recipe for multiple simplex coordinates should not be confused with the entire projection framework. Their general framework permits balance inequalities, and their Section 3 represents equality balances by pairs of opposite inequalities. Thus their network specialization includes (1); here we exploit its underlying equality circulation space. We use the published article and its electronic companion for equation and example numbering.

The shared-simplex gluing principle is also known. In particular, Davarnia [4, Proposition 2.6] proves that convexifying Cartesian components separately is exact when the components share the same simplex. The cited locator belongs to the dissertation, which is distinct from the 2017 journal article. Our block decomposition applies this principle to the independent circulation spaces of an equality-constrained network. We give a direct proof to make the treatment of sparse observations and zero weights explicit.

Several established methods explain the subsequent structural reductions. Reduced reformulation–linearization uses multiplied linear equations to remove common-factor products [13]. In networks, the remaining freedom can be identified with circulations supported on unobserved arcs. The network representation of Cayley hulls of Kis and Horváth [12] gives cut descriptions and transportation

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<sup>1</sup>The published electronic companion is available at [this direct supplement link](#).

projections; their Section 5.7 is a close predecessor for the parallel-path construction. The support geometry of Minkowski sums and edge-generated zonotope refinements is classical [10, 15]. These are ingredients of the graph-specific formulations and separation procedures, rather than new general convexification mechanisms.

Two further comparisons delimit coefficient obstructions. Three-way transportation universality is established by De Loera and Onn [6], and large coefficients in general 0/1 polytopes are classical [2]. A restricted network realization must therefore be specified as part of a coefficient result. Moreover, the integrality of *aggregate* flows on two-terminal series-parallel networks does not give a hull description in selected state-flow coordinates. Almoghrabi et al. [1, Remark 1] explicitly distinguish aggregate integrality from decomposability of the vector of individual commodity flows. Our observed products retain such individual-state information. Neither a sparse extended formulation nor polynomial separation precludes large numerical coefficients after projection.

### 2.3 The known exact disaggregation

For  $\lambda \geq 0$ , define the scaled flow system

$$P(\lambda) = \{f \in \mathbb{R}^E : Af = \lambda b, 0 \leq f \leq \lambda u\}. \quad (6)$$

For  $\lambda > 0$ , this is  $\lambda P$ ; at  $\lambda = 0$  it is  $\{0\}$ , including when  $P$  is empty. Using (6) avoids assigning an ambiguous meaning to  $0\emptyset$ .

**Proposition 2.1** (Simplex-vertex disaggregation). *A point  $(x, y, z)$  belongs to  $H$  if and only if  $y \in \Delta_m$  and there exist state flows  $f^0, \dots, f^m \in \mathbb{R}^E$  such that*

$$f^k \in P(\lambda_k) \quad (k \in \{0, \dots, m\}), \quad \sum_{k=0}^m f^k = x, \quad f_e^j = z_{ej} \quad ((e, j) \in \mathcal{O}). \quad (7)$$

*The equivalence holds even if  $P$  is empty. If the system is feasible, it provides a convex decomposition into at most  $m + 1$  points of  $S$ .*

*Proof.* For a point of  $S$ , set  $f^k = \lambda_k x$ . This satisfies all displayed relations, since  $\sum_k \lambda_k = 1$ . The system is linear in  $(x, y, z, f)$ , so its projection is convex and contains  $H$ .

Conversely, suppose the system is feasible. If  $\lambda_k = 0$ , the finite upper bounds give  $f^k = 0$ . For each positive weight let  $x^k = f^k / \lambda_k \in P$  and define the graph point  $p^k = (x^k, e^k, z^k)$  by  $z_{ej}^k = x_e^k(e^k)_j$ . Its weighted average has  $x$  coordinate  $\sum_k f^k = x$ ,  $y$  coordinate  $\sum_k \lambda_k e^k = y$ , and observed product coordinate  $\lambda_j x_e^j = f_e^j = z_{ej}$  when  $\lambda_j > 0$ . If  $\lambda_j = 0$ , that observed coordinate is zero on both sides. Thus  $(x, y, z) = \sum_{k: \lambda_k > 0} \lambda_k p^k \in H$ . There is at least one positive weight, so feasibility also implies  $P \neq \emptyset$ . This proves the empty-set assertion.  $\square$

The formulation has  $(m + 1)|E|$  state-flow variables before elimination. Its rational size is polynomial in the ordinary network and simplex input, so generic polynomial-time hull optimization and separation already follow from linear programming. The structural issue is avoiding this full state-by-arc expansion, or projecting it into useful original-coordinate inequalities. In particular, eliminating  $f^0 = x - \sum_{j=1}^m f^j$  keeps the residual bounds

$$0 \leq x - \sum_{j=1}^m f^j \leq \left(1 - \sum_{j=1}^m y_j\right)u. \quad (8)$$

These bounds cannot in general be discarded.

### 3 Independent blocks and locally merged states

#### 3.1 Circulation factorization

All graph connectivity terminology below refers to the undirected multigraph underlying  $D$ . Isolated vertices impose only their balance equation. Let  $E_{\text{br}}$  be the bridges. Let  $\mathcal{B}$  be the cyclic blocks: the maximal biconnected edge blocks that contain a cycle, with each self-loop treated as its own block. A pair of parallel edges can form a cycle of length two. The sets  $E_{\text{br}}$  and  $E_B$  for  $B \in \mathcal{B}$  partition  $E$ . The cycle rank of a connected block is

$$r_B = |E_B| - |V_B| + 1. \quad (9)$$

In particular, a loop has rank one. Rank refers to the ambient circulation space, even when capacities reduce the dimension of the feasible polytope.

The equation  $Av = b$  is solvable exactly when the entries of  $b$  sum to zero on every connected component. Necessity follows by summing the incidence rows in a component. For sufficiency, choose a spanning forest, set the nonforest coordinates to zero, and determine each tree-edge flow by the balance sum on one side of its fundamental cut, with the sign determined by the arc orientation. This satisfies every balance equation. This construction uses  $O(|V| + |E|)$  arithmetic operations, and for rational  $b$  produces a rational  $v$  of polynomial encoding length. The reference flow  $v$  need not obey its capacity bounds. If the component condition fails,  $P$  and  $H$  are empty.

Fix such a reference vector  $v$ . For each  $B$ , let  $A_B$  be the incidence matrix restricted to columns in  $E_B$  and rows in  $V_B$ , and define

$$K_B = \{h \in \mathbb{R}^{E_B} : A_B h = 0, -v_{E_B} \leq h \leq u_{E_B} - v_{E_B}\}. \quad (10)$$

Unlike  $P$ , this is a domain of circulation *deviations*; it need not contain the origin.

**Lemma 3.1** (Block factorization). *Every  $g \in \ker A$  vanishes on  $E_{\text{br}}$  and restricts to a circulation on each cyclic block. Conversely, combining arbitrary block circulations and zero bridge coordinates gives an element of  $\ker A$ . Consequently, if every bridge satisfies  $0 \leq v_e \leq u_e$ , the map*

$$x_e = v_e \quad (e \in E_{\text{br}}), \quad x_{E_B} = v_{E_B} + h_B \quad (B \in \mathcal{B}) \quad (11)$$

*is an affine bijection from  $\prod_{B \in \mathcal{B}} K_B$  to  $P$ . If a bridge bound fails,  $P$  is empty. The product of an empty family of block domains is a singleton.*

*Proof.* Choose a spanning forest of the underlying multigraph, omitting loops. Each nonforest edge gives a signed fundamental-cycle circulation: use the unique forest path between its ends, or just the edge itself for a loop. These vectors form a basis of  $\ker A$ . Indeed, subtracting from a circulation the combination that cancels all nonforest coordinates leaves a circulation supported on a forest, which is zero by successive leaf elimination.

Each fundamental cycle lies in a single cyclic block. Hence its vector is zero on every bridge, and on every block it is either zero or a circulation of that block. The basis expansion proves the first assertion. For the converse, extend each block circulation by zero to  $E$ . Its incidence is zero at every vertex, including articulation vertices, so their sum is a global circulation. The sum is direct because the block edge sets are disjoint. Applying this decomposition to  $x - v$  proves that  $Ax = b$  is equivalent to the circulation conditions and the fixed bridge coordinates in (11). The bounds then separate by edge block, giving precisely (10) and the bridge conditions.  $\square$

In particular, an articulation vertex does not carry an additional coupling variable between block deviations. Its fixed net balance is already carried by  $v$ ; every block deviation has zero net balance at that vertex separately. Any spanning tree in  $B$  gives a fundamental-cycle matrix  $C_B \in \{0, \pm 1\}^{E_B \times r_B}$  with chord rows equal to the identity. Thus  $h_B = C_B t_B$  gives unique cycle coordinates  $t_B \in \mathbb{R}^{r_B}$ . These coordinates remain useful when  $K_B$  is lower-dimensional.

### 3.2 The state merger and its exact refinement

For a nonempty convex set  $K$ , use the usual scalar multiplication  $\alpha K = \{\alpha h : h \in K\}$ , so  $0K = \{0\}$ . For nonnegative scalars,

$$\alpha_1 K + \cdots + \alpha_q K = (\alpha_1 + \cdots + \alpha_q) K. \quad (12)$$

If the total weight is positive, the inclusion from left to right follows by dividing by that weight and taking a convex combination; the reverse inclusion uses the same point of  $K$  in each summand. If the total is zero, both sides are  $\{0\}$ . An empty sum is also interpreted as  $\{0\}$ .

For each block define its observed labels and their count by

$$J_B = \{j \in [m] : (e, j) \in \mathcal{O} \text{ for some } e \in E_B\}, \quad a_B = |J_B|. \quad (13)$$

This is a structural definition: a label in  $J_B$  remains present when  $y_j = 0$ . Let  $*_B$  denote a local merged state, distinct from the global residual state 0, and set

$$I_B = J_B \cup \{*_B\}, \quad \lambda_{B,j} = y_j \ (j \in J_B), \quad \lambda_{B,*} = 1 - \sum_{j \in J_B} y_j. \quad (14)$$

The merged state collects all labels absent from  $J_B$  and state 0. Different blocks may merge different global states.

**Theorem 3.2** (Exact blockwise state reduction). *Assume  $P \neq \emptyset$ . A point  $(x, y, z)$  belongs to  $H$  if and only if  $x \in P$ ,  $y \in \Delta_m$ , every observed bridge satisfies*

$$z_{ej} = v_e y_j \quad (e \in E_{\text{br}}, (e, j) \in \mathcal{O}), \quad (15)$$

and for every cyclic block  $B$  there exist vectors  $h^{B,k} \in \mathbb{R}^{E_B}$ ,  $k \in I_B$ , satisfying

$$h^{B,k} \in \lambda_{B,k} K_B \quad (k \in I_B), \quad (16)$$

$$\sum_{k \in I_B} h^{B,k} = x_{E_B} - v_{E_B}, \quad (17)$$

$$h_e^{B,j} = z_{ej} - y_j v_e \quad ((e, j) \in \mathcal{O}, e \in E_B). \quad (18)$$

The block conditions yield state flows in (7), and hence a decomposition into at most  $m + 1$  graph points, by explicit proportional refinement.

*Proof.* By lemma 3.1, every  $K_B$  is nonempty. Start with state flows in (7). Their deviations  $g^k = f^k - \lambda_k v$  are circulations. On  $B$ , they satisfy  $g_{E_B}^k \in \lambda_k K_B$ : division proves this for positive weights, and both deviations and scaled domains are zero for zero weights. For  $j \in J_B$  take  $h^{B,j} = g_{E_B}^j$ , and sum the other deviations into  $h^{B,*}$ . Identity (12) proves (16). Summing the deviations proves (17), and the observed coordinates give (18). On bridges the deviations vanish, proving (15).

Conversely, fix feasible local vectors. For each global state  $k$  and block  $B$ , if  $k \in J_B$ , set  $g_{E_B}^k = h^{B,k}$ . For the remaining states use

$$g_{E_B}^k = \begin{cases} (\lambda_k/\lambda_{B,*})h^{B,*}, & \lambda_{B,*} > 0, \\ 0, & \lambda_{B,*} = 0. \end{cases} \quad (19)$$

If the merged weight is zero, every constituent weight is zero as well. In every case  $g_{E_B}^k \in \lambda_k K_B$ , and the refined vectors sum to the block deviation. Put  $g_e^k = 0$  on bridges and combine blocks. The block factorization gives  $Ag^k = 0$ . Then  $f^k = \lambda_k v + g^k$  satisfies  $Af^k = \lambda_k b$ . Its capacity bounds hold on blocks by the scaled domains, and on bridges because  $P \neq \emptyset$  implies  $0 \leq v_e \leq u_e$ . Equations (17) and (18), together with the bridge equations, give  $\sum_k f^k = x$  and all observed products. Apply [proposition 2.1](#).  $\square$

The domain condition (16) is itself linear:

$$A_B h^{B,k} = 0, \quad -\lambda_{B,k} v_{E_B} \leq h^{B,k} \leq \lambda_{B,k} (u_{E_B} - v_{E_B}). \quad (20)$$

Finite bounds force  $h^{B,k} = 0$  at zero weight. No division by a variable is needed in the formulation; division appears only in the recovery proof at strictly positive weights. If  $J_B = \emptyset$ , the block has one merged state of weight one and its conditions reduce to  $x_{E_B} - v_{E_B} \in K_B$ , already enforced by  $x \in P$ . Thus unobserved blocks require no new variables or constraints.

The theorem is a constructive specialization of the known shared-simplex gluing principle. It does not introduce independent simplex vectors for different blocks: all use the same global  $y$ . It also does not merge observed state slices, whose domains can differ after the products are fixed.

**Example 3.3** (A common matrix is not sufficient for merging). Let  $M = (-1, 1, 2)^\top$ ,  $c^1 = (0, 1, 100)^\top$ , and  $c^2 = (0, 100, 2)^\top$ . Both polytopes  $Q_i = \{q \in \mathbb{R} : Mq \leq c^i\}$  equal  $[0, 1]$ . For weights  $\alpha_1 = \alpha_2 = 1/2$ , the weighted Minkowski sum is still  $[0, 1]$ , whereas the aggregated rows  $Mq \leq (c^1 + c^2)/2$  allow  $q = 2$ . The two upper bounds become  $q \leq 101/2$  and  $2q \leq 51$ . Thus replacing state-specific systems by the sum of their right-hand sides is generally inexact, even for a common matrix and nonempty bounded domains. This is the standard reaggregation boundary discussed by Kis and Horváth [12, Section 2, equation (7)]. The sets in this example are identical, so the set identity (12) still holds. What fails is its replacement by addition of the different, redundant right-hand-side descriptions. In [theorem 3.2](#), every state instead uses the single fixed representation (10), with its bounds scaled by the state weight as in (20). Adding these scaled bounds gives that same representation scaled by the total weight, so the set identity justifies the merged formulation.

Finally, these results concern the hull of (3). Intersecting  $H$  with additional linear side constraints yields a valid relaxation of the graph with those constraints, but need not give its exact convex hull. Such constraints can couple blocks or alter conditional state domains; exactness then requires a separate argument.

## 4 Compression by cycles and observations

This section compresses the exact block formulation in two steps. First, cycle coordinates replace arc coordinates and suppress long paths. Second, observations determine some cycle coordinates, leaving only cycles supported on unobserved arcs. We assume  $P \neq \emptyset$ ; if the original flow domain is empty, its hull is described by any inconsistent linear system.

## 4.1 Fixed coordinates and reduced graphs

The structural parameters can be applied after elementary exact preprocessing. This is useful when capacities force a nominal cycle to carry no freedom.

**Lemma 4.1** (Deleting fixed arcs). *Let  $F \subseteq E$  and suppose  $x_e = c_e$  throughout  $P$  for every  $e \in F$ , where  $c \in \mathbb{Q}^F$  is known. Set  $R = E \setminus F$ ,  $b' = b - A_{:,F}c$ , and*

$$P' = \{x_R : A_{:,R}x_R = b', 0 \leq x_R \leq u_R\}, \quad \mathcal{O}' = \mathcal{O} \cap (R \times [m]).$$

*The hull for  $(P', \mathcal{O}')$ , together with  $x_F = c$  and  $z_{ej} = c_e y_j$  for observed  $e \in F$ , is affinely isomorphic to  $H$ . If  $F$  contains every coordinate fixed throughout  $P$ , then*

$$\text{aff } P' = \{x_R : A_{:,R}x_R = b'\}. \quad (21)$$

*Proof.* Feasible vectors of  $P$  restrict to  $P'$ . Conversely, adjoining  $c$  to a vector of  $P'$  satisfies the original balances and bounds, because  $0 \leq c \leq u_F$ . The same bijection at graph points adjoins the affine coordinates  $z_{ej} = c_e y_j$ , so it commutes with convexification.

For the last assertion, every remaining coordinate is nonconstant on  $P'$ . For each  $e \in R$ , choose two feasible vectors differing in coordinate  $e$ ; their midpoint has  $0 < x_e < u_e$ . Average these midpoints over  $e \in R$ . The resulting feasible vector is strictly inside every remaining bound. A sufficiently small relative neighborhood of that vector in the balance affine space therefore remains feasible, proving (21). If  $R$  is empty, both spaces are the same singleton and the assertion is immediate.  $\square$

The lemma takes the fixed coordinates as input; it does not require a particular detection algorithm. Deleting them can split blocks or reduce their cycle ranks. Once all fixed coordinates are removed, the remaining cycle space is the direction space of the feasible affine hull. Conditional slices at a particular observation or zero state weight can still have smaller dimension.

## 4.2 Suppressing paths

Fix a cyclic block  $B$ . Degrees in this subsection are degrees *within the block*, with loops counted twice. For  $r_B \geq 2$ , suppress maximal paths whose internal vertices have degree two, and give each resulting core edge an arbitrary orientation. A rank-one block is a single cycle; represent its entire cyclic traversal by one core loop. This includes two parallel edges and an original self-loop.

Write  $\mathcal{P}_B$  for the core edges, or equivalently the original paths, and  $k_B = |\mathcal{P}_B|$ . For an original arc  $e$  on path  $p$ , let  $\varepsilon_e \in \{-1, 1\}$  compare its orientation with the chosen path traversal. A circulation has constant signed deviation on a path:

$$x_e = v_e + \varepsilon_e \delta_p \quad (e \in p). \quad (22)$$

The reason is balance of the block circulation at each internal degree-two vertex. Balances at the unsuppressed vertices say precisely that  $\delta$  is a circulation of the core. Arc bounds give the path interval

$$\begin{aligned} L_p &= \max_{e \in p} \ell_e, & U_p &= \min_{e \in p} u'_e, \\ [\ell_e, u'_e] &= \begin{cases} [-v_e, u_e - v_e], & \varepsilon_e = 1, \\ [v_e - u_e, v_e], & \varepsilon_e = -1. \end{cases} \end{aligned} \quad (23)$$

All intervals are finite. Their joint feasibility follows from  $P \neq \emptyset$ .

Suppression preserves cycle rank. If  $r_B \geq 2$ , the connected core has minimum degree at least three: otherwise the original block would have been a single cycle. If its vertex count is  $n_B$ , then  $2k_B \geq 3n_B$  and  $r_B = k_B - n_B + 1$ , so

$$n_B \leq 2r_B - 2, \quad k_B \leq 3r_B - 3. \quad (24)$$

For rank one,  $k_B = 1$ . In particular,  $k_B \leq 3r_B$  in every case.

Choose a spanning tree of the core and let  $\hat{C}_B \in \{0, \pm 1\}^{k_B \times r_B}$  be its fundamental-cycle matrix, normalized to have identity chord rows. Its columns parametrize core circulations bijectively. Thus the feasible core domain is

$$T_B = \{t \in \mathbb{R}^{r_B} : L_p \leq (\hat{C}_B t)_p \leq U_p \quad (p \in \mathcal{P}_B)\}. \quad (25)$$

It is nonempty and bounded, including in lower-dimensional cases. Boundedness follows already from the chord rows. For later use, choose one original representative arc  $e(p)$  on each path and put

$$s_p(x) = \varepsilon_{e(p)}(x_{e(p)} - v_{e(p)}), \quad q_{ej} = \varepsilon_e(z_{ej} - v_e y_j). \quad (26)$$

For  $x \in P$ ,  $s_p(x)$  is its core deviation. A separate observation  $q_{ej}$  is retained for *every* observed original arc, including multiple observations of one path.

**Theorem 4.2** (Cycle-coordinate extended hull). *The hull  $H$  has an exact rational extended formulation with*

$$\sum_{B \in \mathcal{B}} r_B a_B \quad (27)$$

*additional variables and*

$$O \left( |V| + |E| + m + |\mathcal{O}| + \sum_{B: a_B > 0} r_B (a_B + 1) \right) \quad (28)$$

*rows, counting the original domain constraints. Coefficients of  $x$ ,  $z$ , and additional variables can all be chosen from  $\{0, \pm 1\}$ . Coefficients of  $y$  and the right-hand sides may depend on the rational network data. All coefficient encoding lengths are polynomial in the input length.*

*Proof.* Retain  $x \in P$ ,  $y \in \Delta_m$ , and the bridge equations (15). For each  $B$  with  $a_B > 0$ , introduce  $t^{B,j} \in \mathbb{R}^{r_B}$  for  $j \in J_B$  and impose

$$y_j L_p \leq (\hat{C}_B t^{B,j})_p \leq y_j U_p, \quad j \in J_B, \quad p \in \mathcal{P}_B, \quad (29)$$

$$q_{ej} = (\hat{C}_B t^{B,j})_p, \quad (e, j) \in \mathcal{O}, \quad e \in p \subseteq E_B, \quad (30)$$

$$\lambda_{B,*} L_p \leq s_p(x) - \sum_{j \in J_B} (\hat{C}_B t^{B,j})_p \leq \lambda_{B,*} U_p, \quad p \in \mathcal{P}_B. \quad (31)$$

The aggregate deviation  $s(x)$  is a core circulation. Hence the middle quantity in (31) is a core circulation as well. These constraints are exactly [theorem 3.2](#) in core coordinates, with the merged state eliminated by its sum equation. At a zero weight, the finite path bounds force every core path value to zero; the identity chord rows force the entire state vector to zero. Consequently the equivalence includes all simplex boundary strata. Blocks with  $a_B = 0$  add no constraints, as in the block theorem.

There are  $r_B a_B$  variables,  $2k_B(a_B + 1)$  bound rows, and one equation per block observation. Bound (24) proves the row count. Every state variable has coefficient 0, 1, or  $-1$ . An observation row has one original product coordinate, and each residual row uses one original flow coordinate through  $s_p(x)$ ; no sums of different cycle expressions for that flow coordinate are necessary. This proves the stated coefficient bound. Reference flows are sums of rational balances; path endpoints are selected signed capacity offsets. The remaining coefficients are sums of these data, with polynomial rational encoding length.  $\square$

### 4.3 Eliminating observed directions

Define the observed and unobserved arcs for a block and label by

$$O_{Bj} = \{e \in E_B : (e, j) \in \mathcal{O}\}, \quad U_{Bj} = E_B \setminus O_{Bj}.$$

Let  $c(V_B, U_{Bj})$  count all connected components of this unobserved undirected subgraph, including isolated vertices. Its cycle rank is

$$\rho_{Bj} = |U_{Bj}| - |V_B| + c(V_B, U_{Bj}). \quad (32)$$

Loops and parallel edges keep their usual multigraph meanings. In particular, each unobserved loop contributes one to the cycle rank.

**Lemma 4.3** (Unit minors for cycle elimination). *A fundamental-cycle matrix with identity chord rows is totally unimodular. Let  $C$  be such a matrix, let  $D$  consist of  $d$  linearly independent rows, and choose  $d$  columns  $I$  such that  $B = D_{:,I}$  is nonsingular. Write  $F$  for the complementary columns. Then, for every row  $c$  of  $C$ , the vectors*

$$W = c_I B^{-1}, \quad R = c_F - W D_{:,F} \quad (33)$$

*have entries in  $\{0, \pm 1\}$ .*

*Proof.* Delete one incidence row of the connected core and partition the reduced incidence matrix as  $[N_T \ N_Q]$  into tree and chord columns. A tree basis has determinant  $\pm 1$ . Every square incidence minor is 0 or  $\pm 1$ : induction expands a column with at most one nonzero entry; if every column has two entries, they are opposite and the row sum is zero. Every minor of  $N_T^{-1} N_Q$  equals, up to sign, the determinant of a matrix obtained by replacing the corresponding tree columns by chord columns, divided by  $\det N_T$ . This follows by multiplying the replacement matrix by  $N_T^{-1}$  and expanding its unchanged identity columns. The replacement is a square incidence submatrix. Therefore  $-N_T^{-1} N_Q$  is totally unimodular. Appending identity chord rows preserves this property, proving the assertion for  $C$ . A core consisting of one loop has  $C = (1)$  directly.

In particular  $\det B = \pm 1$ . Cramer's rule applied to  $WB = c_I$  expresses each entry of  $W$  as a selected-row replacement minor of  $C$  divided by  $\det B$ . For a free column  $f$ , the bordered determinant identity gives

$$R_f = \frac{\det \begin{pmatrix} B & D_{:,f} \\ c_I & c_f \end{pmatrix}}{\det B}.$$

The numerator is a minor of  $C$ , or zero when its rows repeat. These formulas prove (33). Empty pivot or free sets have their usual zero-dimensional matrix interpretation.  $\square$

**Theorem 4.4** (Observation-sensitive extended hull). *There is an exact rational extended formulation for  $H$  with*

$$\sum_{B \in \mathcal{B}} \sum_{j \in J_B} \rho_{Bj} \quad (34)$$

*additional variables and the row bound (28). Coefficients of  $x$ ,  $z$ , and additional variables belong to  $\{0, \pm 1\}$ ; simplex coefficients and right-hand sides have polynomial rational encoding length. The variable count is an upper bound for this construction, not a lower bound for arbitrary extended formulations.*

*Proof.* Fix  $B$  and  $j \in J_B$ , suppress their subscripts, and write  $C = \widehat{C}_B$ ,  $r = r_B$ . Choose a maximal independent set of observed path rows of  $C$ , giving  $D \in \mathbb{R}^{d \times r}$ . For each selected row choose one actual observed arc and use its value from (26), giving  $q \in \mathbb{R}^d$ . The selected entries of  $q$  involve distinct original product variables. All other observations will remain as equations, including any repeated path values.

Choose pivot columns  $I$  and complementary columns  $F$  as in lemma 4.3. Retain  $g = t_F \in \mathbb{R}^{r-d}$  and eliminate the pivot coordinates by the reversible substitution

$$t_I = B^{-1}(q - D_{:,F}g). \quad (35)$$

For every core path  $p$  with row  $c_p$  define

$$\phi_{pj} = (c_{p,I}B^{-1})q + (c_{p,F} - c_{p,I}B^{-1}D_{:,F})g. \quad (36)$$

Replace  $(\widehat{C}_B t^{B,j})_p$  by  $\phi_{pj}$  in (29)–(31). Selected observation equations become identities and can be omitted; all other observation equations remain. Formula (35) proves equivalence in both directions, so no projection condition is lost.

By lemma 4.3, each selected product and retained variable has coefficient 0, 1, or  $-1$  in every  $\phi_{pj}$ . A nonselected observation adds a distinct product coordinate; it cannot add a second occurrence of a selected product. Different labels use disjoint product and auxiliary columns. The residual uses one original flow coordinate through  $s_p(x)$ . Thus collecting terms does not increase any  $x$ ,  $z$ , or auxiliary coefficient beyond one in magnitude. Only coefficients of  $y$  collect reference-flow and interval data. Their encoding lengths remain polynomial. Elimination adds no rows.

It remains to identify  $r - d$ . Restrict a block circulation to  $O_{Bj}$ . The kernel is precisely the space of circulations supported on  $U_{Bj}$ , whose dimension is (32) by the incidence-rank formula. The core matrix parametrizes all block circulations, and observing any arc of a suppressed path fixes its signed path value. Therefore the rank of the restriction map is exactly the rank  $d$  of the observed core rows. Rank-nullity gives  $r_B - d = \rho_{Bj}$ , proving (34).  $\square$

For clarity, the final block inequalities are

$$y_j L_p \leq \phi_{pj} \leq y_j U_p, \quad j \in J_B, p \in \mathcal{P}_B, \quad (37)$$

$$q_{ej} = \phi_{pj}, \quad (e, j) \in \mathcal{O}, e \in p \subseteq E_B, \quad (38)$$

$$\lambda_{B,*} L_p \leq s_p(x) - \sum_{j \in J_B} \phi_{pj} \leq \lambda_{B,*} U_p, \quad p \in \mathcal{P}_B. \quad (39)$$

The original flow and simplex domains and bridge observations remain part of the formulation. In particular, reconstructing state products by balance equations alone is insufficient: all displayed state and residual bounds must also be imposed.

The row counts do not bound matrix nonzeros when ranks grow. If  $N = |V| + |E| + m + |\mathcal{O}|$  and  $\mathcal{O}_B = \mathcal{O} \cap (E_B \times [m])$ , both constructions have the explicit nonzero bound

$$O \left( N + \sum_{B: a_B > 0} [r_B^2(a_B + 1) + r_B |\mathcal{O}_B|] \right). \quad (40)$$

Indeed, a reconstructed path has  $O(r_B)$  selected-product and auxiliary terms per label; there are  $O(r_B)$  paths and  $a_B$  explicit labels. Each remaining observation also uses  $O(r_B)$  terms. Hence bounded maximum block rank gives size linear in the sparse input, even as  $m$  and the graph grow. The whole formulation need not be totally unimodular. The unit coefficient statements use the displayed rational row scaling; clearing denominators in the simplex coefficients can destroy that unit bound in a primitive integer normalization.

#### 4.4 Forest complements and product completion

**Corollary 4.5** (An original-coordinate hull on arbitrary graphs). *Suppose  $(V_B, U_{Bj})$  is a forest for every  $B$  and  $j \in J_B$ . Then (37)–(39), the original domains, and the bridge equations describe  $H$  with no additional variables. They have unit flow and product coefficients and the row bound (28).*

*Proof.* An undirected multigraph is a forest exactly when its cycle rank is zero. Thus every  $\rho_{Bj}$  in (34) vanishes. Apply theorem 4.4.  $\square$

Equivalently, the observations for each locally observed label meet every cycle of its block. Entirely unobserved labels impose no condition: they are already included in the merged state. The condition is sufficient; it does not assert that remaining unobserved cycles prevent an original-coordinate description with unit coefficients.

**Proposition 4.6** (Minimum individual-coordinate completion). *Fix  $B$  and  $j \in J_B$ . Among additions of individual missing arc-product coordinates, exactly  $\rho_{Bj}$  are necessary and sufficient to determine the full state circulation from the observed coordinates and linear balance equations on the ambient circulation space. A minimum choice consists of the nonforest edges of any spanning forest of  $(V_B, U_{Bj})$ . The additional variables in theorem 4.4 may instead be chosen to represent these individual missing products, with the same variable and row bounds and unit flow, product, and auxiliary coefficients.*

*Proof.* The existing restriction map has nullity  $\rho_{Bj}$ . Each additional scalar coordinate can raise its rank by at most one, proving necessity. Let  $F_{Bj}$  be a spanning forest of the unobserved subgraph, including all its vertices. Add observations on  $U_{Bj} \setminus F_{Bj}$ , which has  $\rho_{Bj}$  edges. The remaining unobserved graph is a forest, so the restriction kernel is zero by the proof of theorem 4.4. This proves sufficiency and the stated minimum.

Apply corollary 4.5 to the completed observation set and regard the added product coordinates as auxiliary variables. Projection onto the original observations commutes with convexification, so this is an extended formulation for the original  $H$ . Completing labels only in  $J_B$  does not introduce new labels. The added equations number  $\sum_{B,j} \rho_{Bj} \leq \sum_B r_B a_B$ , already absorbed by the row bound. The corollary's coefficient property includes the added coordinates.  $\square$

There is a direct interpretation of this completion. The block state flows satisfy  $A_B f^{B,j} = y_j A_B v_{E_B}$ , since their deviations are block circulations. Once the nonforest state products are known, sum these equations on either side of a cut formed by deleting a remaining forest edge from its component. That edge is the only unknown crossing edge, so its state product is an affine expression

in the known state products and the state weight, with unit product coefficients. Component balance equations enforce consistency if the unobserved forest is disconnected. This is network elimination of the kind underlying reduced RLT: Liberti and Pantelides [13, Theorem 3.1 and Section 2] reconstruct common-factor products by multiplying full-row-rank linear equations and retaining products on suitable complementary coordinates. Here the extra information is the observation-complement graph criterion and its exact sparse simplex-hull consequence.

The minimum in [proposition 4.6](#) concerns reconstruction by *individual coordinates on the ambient linear space*. It is neither an extension-complexity lower bound nor a claim about the number of coordinates needed on a capacity-degenerate slice. In particular,  $\lambda_j = 0$  forces every state product to zero. The preprocessing in [lemma 4.1](#) can remove fixed arcs before the parameters are computed, but it does not remove this conditional-slice distinction.

All formulations also give explicit recovery. From retained coordinates, recover each  $t^{B,j}$  using (35), reconstruct path deviations by  $\hat{C}_B t^{B,j}$ , and subtract their sum from  $s(x)$  to recover the merged state. Expand deviations along the original paths using their signs. The proportional refinement (19) then supplies the global state flows and the at-most- $m + 1$ -point decomposition from [proposition 2.1](#). This recovery is exact rational arithmetic for rational input and a rational feasible extension; merely solving the formulation numerically does not provide an exact certificate.

**Example 4.7** (Three observations on a  $K_4$  block). Orient every edge of  $K_4$  from the smaller to the larger vertex and take unit capacities and unit flow from vertex 1 to vertex 4. Thus  $b = (-1, 0, 0, 1)^\top$ . Use the reference flow  $v_{14} = 1$  and zero on the other arcs. For one label  $j$ , observe 13, 14, and 24; the unobserved edges 12, 23, and 34 form a spanning tree. Writing  $q_{13} = z_{13,j}$ ,  $q_{14} = z_{14,j} - y_j$ , and  $q_{24} = z_{24,j}$ , balance gives the remaining state deviations

$$h_{12} = -q_{13} - q_{14}, \quad h_{23} = -q_{13} - q_{14} - q_{24}, \quad h_{34} = -q_{14} - q_{24}.$$

For example, nonnegativity of the reconstructed flow on 23 gives

$$z_{13,j} + z_{14,j} + z_{24,j} \leq y_j. \tag{41}$$

This cut is absent from independent product convexification. To see the gap, take  $m = 1$ ,  $y_1 = 1/2$ ,  $z_{13,1} = z_{14,1} = z_{24,1} = 1/5$ , and let  $x$  average the four paths 1234, 124, 134, and 14 equally. Then  $x_{12} = x_{34} = 1/2$  and the other four arc flows are  $1/4$ . All three observed products satisfy (5), but the left side of (41) is  $3/5 > 1/2$ . The complete hull requires all reconstructed bounds, including the residual bounds, not only this one cut.

## 5 Explicit hulls and separation on structured blocks

The extended formulations in [section 4](#) apply to arbitrary graphs. On some block classes, their state coordinates can be projected out explicitly while preserving unit flow and product coefficients. We first give a linear-arithmetic oracle for cycles and theta blocks, then a transportation oracle for any number of parallel paths. All graph classes refer to the underlying multigraph: orientations remain arbitrary, and different blocks may meet at articulation vertices. Bridges retain (15); independent loops use the scalar cycle formulas.

### 5.1 Cycle and theta state domains

A theta block consists of three internally vertex-disjoint paths between two terminals. Orient their traversals from the first terminal to the second. The three circulation deviations sum to zero, so

write them as  $s, t, -s - t$ . Put  $h(s, t) = s + t$  and absorb the sign of the third path into  $\sigma_e$ : on the first two paths  $\sigma_e = \varepsilon_e$ , and on the third  $\sigma_e = -\varepsilon_e$ . Thus

$$x_e = v_e + \sigma_e \eta_e(s, t), \quad \eta_e \in \{s, t, h\}, \quad h = s + t. \quad (42)$$

The symbol  $h$  here is a linear coordinate form, not the number of states. Signed interval intersection as in (23) gives

$$T_B = \{(s, t) : L_s \leq s \leq U_s, L_t \leq t \leq U_t, L_h \leq s + t \leq U_h\}. \quad (43)$$

A cycle has the analogous scalar form  $x_e = v_e + \sigma_e s$  and base interval  $[L_s, U_s]$ . Read the aggregate coordinates  $s(x), t(x)$  from one representative arc on each of the first two paths; then  $h(x) = s(x) + t(x)$ . These representative arcs are distinct.

Fix a block and use its local states  $I_B$  and weights from (14). Suppress  $B$  in the local formulas below, so  $\lambda_j = \lambda_{B,j}$ , including the merged state  $*$ . Define  $\tilde{q}_{ej} = \sigma_e(z_{ej} - v_e y_j)$  for observations in this block. Initialize and tighten each coordinate interval by

$$\ell_{\eta j} = \max(\{\lambda_j L_\eta\} \cup \{\tilde{q}_{ej} : (e, j) \in \mathcal{O}, \eta_e = \eta\}), \quad u_{\eta j} = \min(\{\lambda_j U_\eta\} \cup \{\tilde{q}_{ej} : (e, j) \in \mathcal{O}, \eta_e = \eta\}). \quad (44)$$

For  $*$  there are no observations. Multiple observations on a path must agree: they all enter both endpoint lists. At zero weight the initial interval is the singleton zero, so feasibility forces every observed coordinate and every state deviation to be zero.

The cycle state domain is  $[\ell_{sj}, u_{sj}]$ . The theta state domain is

$$Q_j = \{(s, t) : \ell_{sj} \leq s \leq u_{sj}, \ell_{tj} \leq t \leq u_{tj}, \ell_{hj} \leq s + t \leq u_{hj}\}. \quad (45)$$

By theorem 3.2, block membership asks whether the aggregate coordinate belongs to the Minkowski sum of these local state domains. This reduction holds at every simplex boundary and for lower-dimensional state domains.

**Lemma 5.1** (Theta feasibility and tight supports). *The state domain  $Q_j$  is nonempty if and only if*

$$\ell_{\eta j} \leq u_{\eta j} \quad (\eta = s, t, h), \quad \ell_{sj} + \ell_{tj} \leq u_{hj}, \quad \ell_{hj} \leq u_{sj} + u_{tj}. \quad (46)$$

*When it is nonempty, its attained minimum and maximum of each coordinate form are*

$$\begin{aligned} a_{sj} &= \max\{\ell_{sj}, \ell_{hj} - u_{tj}\}, & b_{sj} &= \min\{u_{sj}, u_{hj} - \ell_{tj}\}, \\ a_{tj} &= \max\{\ell_{tj}, \ell_{hj} - u_{sj}\}, & b_{tj} &= \min\{u_{tj}, u_{hj} - \ell_{sj}\}, \\ a_{hj} &= \max\{\ell_{hj}, \ell_{sj} + \ell_{tj}\}, & b_{hj} &= \min\{u_{hj}, u_{sj} + u_{tj}\}. \end{aligned} \quad (47)$$

*Proof.* The rectangle of individual coordinate intervals is nonempty exactly when its two intervals are nonempty. Its attainable sums fill  $[\ell_{sj} + \ell_{tj}, u_{sj} + u_{tj}]$ . Requiring that interval to intersect the nonempty sum interval gives precisely (46). For a fixed  $s$ , a feasible  $t$  exists if and only if  $s \in [\ell_{sj}, u_{sj}]$  and  $s \in [\ell_{hj} - u_{tj}, u_{hj} - \ell_{tj}]$ . This proves both tight  $s$  bounds and their attainment. Interchanging  $s, t$  proves the next pair, and intersecting the two attainable sum intervals proves the last.  $\square$

**Lemma 5.2** (Six supports suffice for planar sums). *For any finite family of nonempty domains of the form (45), including segments and points,*

$$\sum_j Q_j = \left\{ (s, t) : \sum_j a_{\eta j} \leq \eta(s, t) \leq \sum_j b_{\eta j} \quad (\eta = s, t, h) \right\}. \quad (48)$$

*The empty sum is the singleton zero, represented by six zero supports.*

*Proof.* Let  $\mathcal{N} = \{\pm(1,0), \pm(0,1), \pm(1,1)\}$ . Every edge of a two-dimensional  $Q_j$  has a normal in  $\mathcal{N}$ , because  $Q_j$  is defined by inequalities with those normals. A one-dimensional  $Q_j$  also has a direction perpendicular to a member of  $\mathcal{N}$ : at a relative interior point of the segment, at least one defining inequality is tight and its boundary contains the segment. Otherwise that point would have a two-dimensional neighborhood in  $Q_j$ .

For a linear objective with normal  $n$ , its maximizing face over a sum is the sum of its maximizing faces over the summands. Indeed, objective values add, and equality in the sum of the separate maximum values holds exactly when every summand attains its maximum. If the sum is two-dimensional, each exposed edge is therefore a sum of points and parallel segments, with at least one nontrivial segment. Its direction is a direction already occurring in a summand. Thus every edge normal of the sum lies in  $\mathcal{N}$ , and its support values are the sums of those in [lemma 5.1](#). These supports describe the polygon.

If the sum is a nontrivial segment, each nonpoint summand is a segment parallel to it. Its direction is one of the three permitted directions. The two normals perpendicular to that direction enforce its supporting line, and at least one of the other coordinate forms varies along that line. The lower and upper supports of that form give exactly the segment endpoints. If the sum is a point, the supports of  $s$  and  $t$  fix it. This proves the same six-inequality description in all dimensions.  $\square$

The planar restriction matters. In higher dimension, an exposed facet can be a sum of lower-dimensional faces with different directions, so the preceding edge argument does not give a fixed list of facet normals. The support and normal-fan identities used here are classical [\[10\]](#); the explicit formulas exploit these particular block domains.

**Theorem 5.3** (Cycle/theta original-coordinate hull). *Suppose every cyclic block is a cycle or a theta block. The hull  $H$  is described by the original flow and simplex constraints, the bridge equations, the cycle conditions*

$$\ell_{sj} \leq u_{sj} \quad (j \in I_B), \quad \sum_{j \in I_B} \ell_{sj} \leq s(x) \leq \sum_{j \in I_B} u_{sj}, \quad (49)$$

and, for every theta block, the local conditions [\(46\)](#) and six aggregate conditions

$$\sum_{j \in I_B} a_{\eta j} \leq \eta(s(x), t(x)) \leq \sum_{j \in I_B} b_{\eta j} \quad (\eta = s, t, h). \quad (50)$$

All endpoints are evaluated by [\(44\)](#). These expressions define a finite family of linear inequalities with coefficients in  $\{0, \pm 1\}$  on  $x, z$ .

*Proof.* For intervals, nonemptiness is checked separately, and their Minkowski sum has the summed lower and upper endpoints. The theta case follows from [lemmas 5.1](#) and [5.2](#). The exact equivalence with the global hull then follows from [theorem 3.2](#).

Every lower endpoint in [\(44\)](#) is a maximum of affine functions, and every upper endpoint is a minimum. These properties also hold for the lower and upper supports in [\(47\)](#): for example,  $\ell_{hj} - u_{tj}$  is a maximum of affine differences. Consequently each displayed condition can be written as a maximum of finitely many affine functions bounded above by zero. Requiring every affine branch to be at most zero is an equivalent finite linear description.

In a support branch, a state's product contribution uses either one coordinate type or two distinct types. Hence each observed product occurs at most once. Different states have different product indices. A local same-type interval comparison either uses two distinct observations or cancels an identical one; other local comparisons use distinct types. The aggregate  $s(x), t(x)$  use separate representative arcs, and  $h(x)$  uses their sum. Thus every flow or product coefficient is 0, 1, or  $-1$ . Simplex coefficients contain the network data. The assertion concerns this rational row scaling, with the normalization qualifications given after [\(40\)](#).  $\square$

## 5.2 A linear-arithmetic oracle and compact recovery

If any condition in [theorem 5.3](#) fails, select a maximizing affine branch of its convex piecewise-affine left side after moving all terms to one side. That branch is bounded above by zero at every hull point and has the same positive violation at the candidate. Thus endpoint and support comparisons produce a globally valid linear cut directly; they do not merely produce a local linear approximation. Ties may be resolved arbitrarily among active branches.

There are  $a_B + 1$  local states in block  $B$ , and  $\sum_B a_B \leq |\mathcal{O}|$ . A constant number of endpoint and support operations suffices for each cycle/theta state after its observations have been scanned. Checking the original domains and all block conditions therefore uses

$$O(|V| + |E| + m + |\mathcal{O}|) \quad (51)$$

rational arithmetic operations, given observation indices grouped by block and label. Graph blocks, paths, and reference flows can be preprocessed in linear time. Sorting an ungrouped observation list adds  $O(|\mathcal{O}| \log |\mathcal{O}|)$  comparisons if required. These are arithmetic and comparison counts, not constant-bit-cost claims. All evaluated rational expressions and emitted cut coefficients have polynomial encoding length.

**Proposition 5.4** (Constructive cycle/theta recovery). *For a rational hull point in the scope of [theorem 5.3](#), a compact rational decomposition into at most  $m+1$  graph points can be computed within (51), with the same grouping proviso. The compact representation stores the global weights, reference flow, one default coordinate per block, and exceptions for observed block/label pairs. Expanding all flow vectors can require  $\Theta((m+1)|E|)$  output entries.*

*Proof.* Order the local states in a theta block and precompute suffix sums of the six tight supports. Suppose the remaining aggregate is  $r = (r_s, r_t)$ , with  $r_h = r_s + r_t$ , and the next state is  $i$ . Its permissible coordinate lies in  $Q_i \cap (r - \sum_{j>i} Q_j)$ . By [lemma 5.2](#), this is another theta polygon with intervals

$$\max \left\{ \ell_{\eta i}, r_{\eta} - \sum_{j>i} b_{\eta j} \right\} \leq \eta \leq \min \left\{ u_{\eta i}, r_{\eta} - \sum_{j>i} a_{\eta j} \right\}, \quad \eta = s, t, h. \quad (52)$$

It is nonempty because the remaining aggregate belongs to the remaining sum. In any nonempty theta polygon with interval endpoints  $\ell, u$ , choose

$$s = \max\{\ell_s, \ell_h - u_t\}, \quad t = \max\{\ell_t, \ell_h - s\}. \quad (53)$$

By [lemma 5.1](#), the chosen  $s$  lies between its tight bounds, in particular  $s \leq u_s$  and  $s \leq u_h - \ell_t$ . Both arguments defining  $t$  are at most  $u_t$  because  $s \geq \ell_h - u_t$ . The sum is at least  $\ell_h$ ; if  $t = \ell_t$  it is at most  $u_h$ , and otherwise it equals  $\ell_h$ . Thus the chosen point is feasible. Subtract it from  $r$  and continue. Empty suffixes use zero supports. The scalar interval construction is the same intersection-and-subtraction procedure in one dimension.

Normalize each positive-weight explicit state coordinate by its weight. Normalize a positive-weight merged coordinate by its total weight and store it as the block's default for all constituent global states. If that weight is zero, the default is unused and may be the aggregate block coordinate, which is feasible because  $x \in P$ . An unobserved block also uses its aggregate coordinate as the default. The refinement proof of [theorem 3.2](#) combines these coordinates into feasible global flows with weights  $\lambda_0, \dots, \lambda_m$ ; zero weights are omitted. The number of exceptions is at most  $|\mathcal{O}|$ , and all operations and stored coordinates obey the claimed bound. Enumerating a dense flow for every positive state is a separate output cost.  $\square$

**Example 5.5** (Separate state hulls miss a joint restriction). Consider three parallel arcs of unit capacity carrying unit total flow. Let their first two aggregate flows be  $X_s = X_t = 1/3$ , and let  $y_1 = y_2 = 1/3$ . Observe both first-two-arc products for both labels, and set all four to zero. For either label alone, the weighted state flow  $(0, 0, 1/3)$  and its complementary flow  $(1/3, 1/3, 0)$  give a feasible one-label hull decomposition. Thus the candidate belongs to the intersection of the two separately convexified one-label hulls.

Jointly, the two observed states would consume  $2/3$  of the third arc, whose aggregate flow is only  $1/3$ . The violated joint inequality is

$$z_{s1} + z_{t1} + z_{s2} + z_{t2} \geq X_s + X_t - (1 - y_1 - y_2). \quad (54)$$

Its violation is  $1/3$ . It follows from the residual state's upper bound on the sum of the first two flows, and is included in (50). The example illustrates the known joint disaggregation mechanism rather than a separate convexification principle.

### 5.3 Any number of parallel paths

Now suppose each cyclic block, apart from independent loops, consists of  $k \geq 2$  internally vertex-disjoint paths between two terminals. The number  $k$  is unrestricted. Orient all path traversals from the first terminal to the second, and use the signs  $\varepsilon_e$  of (22). The core domain is

$$L_i \leq \delta_i \leq U_i \quad (i \in [k]), \quad \sum_{i=1}^k \delta_i = 0. \quad (55)$$

Again write  $\lambda_j = \lambda_{B,j}$  for local states, including  $*$ . For each local state  $j \in I_B$ , intersect its scaled path bounds with all observations on that path:

$$\ell_{ij} = \max(\{\lambda_j L_i\} \cup \{q_{ej} : (e, j) \in \mathcal{O}, e \in i\}), \quad u_{ij} = \min(\{\lambda_j U_i\} \cup \{q_{ej} : (e, j) \in \mathcal{O}, e \in i\}). \quad (56)$$

Here  $q_{ej}$  uses (26), without the theta-specific sign change. Let  $\bar{\delta}_i = s_i(x)$  be the aggregate path deviations. Exact block membership is equivalent to finding a matrix  $w = (w_{ij})$  such that

$$\ell_{ij} \leq w_{ij} \leq u_{ij}, \quad \sum_i w_{ij} = 0 \quad (j \in I_B), \quad \sum_j w_{ij} = \bar{\delta}_i \quad (i \in [k]). \quad (57)$$

This follows directly from [theorem 3.2](#).

**Theorem 5.6** (Complete parallel-path subset description). *In this graph class,  $H$  is described by the original domains, bridge equations, scalar loop conditions, and the following conditions for every parallel-path block:*

$$\ell_{ij} \leq u_{ij} \quad (i, j), \quad \sum_i \ell_{ij} \leq 0 \leq \sum_i u_{ij} \quad (j \in I_B), \quad (58)$$

and, for every subset  $S \subseteq [k]$ ,

$$\sum_{i \in S} \bar{\delta}_i \leq \sum_{j \in I_B} \min \left\{ \sum_{i \in S} u_{ij}, - \sum_{i \notin S} \ell_{ij} \right\}. \quad (59)$$

After expanding affine branches, this is a finite linear description whose coefficients on  $x, z$  lie in  $\{0, \pm 1\}$ , with the same rational-scaling qualification as [theorem 5.3](#).

*Proof.* The local conditions are necessary for (57). In a zero-sum column, the sum on  $S$  is bounded by its upper endpoints on  $S$  and by the negative lower endpoints on its complement. Summing over columns proves (59).

For sufficiency, shift all entries by their lower endpoints and define

$$g_{ij} = w_{ij} - \ell_{ij}, \quad C_{ij} = u_{ij} - \ell_{ij}, \quad r_i = \bar{\delta}_i - \sum_j \ell_{ij}, \quad c_j = -\sum_i \ell_{ij}. \quad (60)$$

We seek a nonnegative matrix  $g \leq C$  with row sums  $r$  and column sums  $c$ . The local conditions give  $C_{ij} \geq 0$  and  $c_j \geq 0$ . For each  $i$ , apply (59) to  $[k] \setminus \{i\}$ . Since  $\sum_i \bar{\delta}_i = 0$ , this gives

$$-\bar{\delta}_i \leq \sum_j \min \left\{ \sum_{h \neq i} u_{hj}, -\ell_{ij} \right\} \leq -\sum_j \ell_{ij},$$

so  $r_i \geq 0$  as well. Moreover  $R := \sum_i r_i = \sum_j c_j$ .

Build a directed network with a source, one node for each row, one node for each column, and a sink. Source-to-row capacities are  $r_i$ , row-to-column capacities are  $C_{ij}$ , and column-to-sink capacities are  $c_j$ . A flow of value  $R$  saturates all source and sink arcs and yields exactly the desired matrix. Fixing the source-side row set  $S$ , the minimum cut over the side choices of column nodes has capacity

$$\sum_{i \notin S} r_i + \sum_j \min \left\{ \sum_{i \in S} C_{ij}, c_j \right\}. \quad (61)$$

Indeed, a sink-side column pays its incoming capacities from  $S$ , while a source-side column pays  $c_j$ ; the choices are independent. This cut has capacity at least  $R$  if and only if

$$\sum_{i \in S} r_i \leq \sum_j \min \left\{ \sum_{i \in S} C_{ij}, c_j \right\}.$$

Adding  $\sum_{i \in S, j} \ell_{ij}$  to both sides converts this inequality exactly into (59). Hence every cut has capacity at least  $R$ . The maximum-flow/minimum-cut theorem [8] gives the desired flow, including the case  $R = 0$ , when all targets vanish. Undo the shift to obtain (57), then refine local states and combine blocks by theorem 3.2.

The right side of (59) is a minimum of finitely many affine expressions:  $u_{ij}$  and  $-\ell_{ij}$  have this property, and it is preserved by sums and minima. Branch expansion therefore gives an equivalent finite linear family. In a branch for one state, either one upper endpoint from each path in  $S$  is used, or one negated lower endpoint from each complementary path is used. Each observation belongs to only one path and one state, so its coefficient has magnitude at most one. The local conditions have the same property, with possible cancellation of a repeated observation in a same-interval comparison. The aggregate left side uses distinct representative arcs. This proves the coefficient assertion.  $\square$

For  $k = 3$ , the six nonempty proper subsets are the supports in (47), after setting  $(\delta_1, \delta_2, \delta_3) = (s, t, -h)$ . The empty and full subsets are redundant under (58). For larger  $k$ , the subset family may be exponential. Its completeness comes from the transportation projection, not from an extension of the planar edge argument. This is an application of classical transportation feasibility [9]; the network-Cayley projection of Kis and Horváth [12, Section 5.7] is a direct predecessor.

## 5.4 Separation by flow and reconstruction

The theorem supplies an oracle without enumerating the subsets. Check the original domains, bridge equations, and local conditions first. If a shifted row target  $r_i$  in (60) is negative, the valid condition

$$\bar{\delta}_i \geq \sum_j \ell_{ij} \quad (62)$$

separates the point; its validity was proved from the complementary subset above. Otherwise all transportation capacities and targets are nonnegative. Compute a maximum flow. Value  $R$  proves block membership and returns its state matrix. A smaller value gives a minimum cut; its source-side row set violates (59), because minimizing column sides can only decrease the cut capacity.

To return an original-coordinate linear inequality, select the active minimum alternatives and active affine endpoint expressions in the violated subset condition. Their sum is an affine majorant of its concave right side and agrees with it at the candidate. Hence the resulting affine upper bound is globally valid and strictly violated. Local violations and (62) are handled by their active affine branches in the same way.

For a block with  $k_B$  paths and  $a_B$  observed labels, the transportation network has

$$k_B + a_B + 3 \text{ nodes}, \quad k_B(a_B + 1) + k_B + a_B + 1 \text{ arcs.} \quad (63)$$

Graph and endpoint preprocessing, target construction, and cut extraction use

$$O\left(|V| + |E| + m + |\mathcal{O}| + \sum_B k_B(a_B + 1)\right) \quad (64)$$

arithmetic operations after observation grouping, followed by one maximum flow computation per parallel-path block. A polynomial-time algorithm such as shortest augmenting paths [7] gives polynomial bit complexity for rational data. One way to see the encoding bound is to clear the finitely many capacity denominators with a common multiple; its encoding length is at most their summed encoding lengths. Flow updates then use integers bounded by the total capacity. An arbitrary augmenting-path rule is not needed for this complexity claim. When  $k_B$  grows, (64) and the flow solves replace the linear cycle/theta bound.

The returned matrices provide exact rational block state deviations. Normalize positive-weight columns and store the normalized merged column as the default for its constituent global states. Zero-weight columns vanish by their original scaled path bounds, and no division by zero occurs. Together with the reference flow and block paths, the resulting compact decomposition has storage of order (64); expanding every dense state-flow vector can still require  $\Theta((m + 1)|E|)$  entries.

The cycle rank of a parallel-path block is  $k_B - 1$ , so this theorem permits unbounded rank within a block. It does not cover every series-parallel block. In particular, several serial parallel gadgets inside one outer bypass can have interacting internal state profiles that are not represented by the single transportation matrix (57). The present results supply explicit structural oracles and unit-coefficient descriptions; generic polynomial separation already follows from [proposition 2.1](#).

## 6 Separation and coefficient bounds at bounded block rank

The preceding unit-coefficient oracles rely on special block shapes. We now allow arbitrary blocks and bound their cycle ranks instead. The construction uses finite libraries determined by each block's signed cycle normals. It has no online linear-programming subproblem. Its parameter

dependence is large, so it provides a structural bound rather than a runtime advantage over the compressed formulations.

For  $s \geq 1$  define

$$H_1 = 1, \quad H_s = \max\{1, \lfloor (s-1)^{(s-1)/2} \rfloor\} \quad (s \geq 2). \quad (65)$$

The first four values are 1, 1, 2, 5. Put  $r = \max_{B \in \mathcal{B}} r_B$ . If there are no cyclic blocks, the flow and simplex domains and bridge equations already describe the hull, so no rank-dependent procedure is needed.

## 6.1 A fixed normal matrix for every state

Fix a cyclic block of rank  $s$ . Use original chord arcs as cycle coordinates, with the fundamental-cycle matrix  $C \in \{0, \pm 1\}^{E_B \times s}$  of lemma 3.1, so  $x_{E_B} = v_{E_B} + C\theta$  and the chord rows are the identity. Let  $\mathcal{A}$  contain the distinct nonzero rows of  $C$  with both signs, and let  $M$  be the matrix with row set  $\mathcal{A}$ . It is totally unimodular by lemma 4.3, has rank  $s$ , contains  $\pm e_i$ , and has at most  $3^s - 1$  rows. Repeated normals are combined by defining

$$\beta_a = \min(\{u_e - v_e : C_e = a\} \cup \{v_e : -C_e = a\}), \quad T = \{\theta : M\theta \leq \beta\}. \quad (66)$$

The lists in this formula are nonempty by construction. Every arc in a cyclic block has a nonzero cycle row, since it lies on a cycle. Hence these inequalities include all arc bounds. The domain  $T$  is bounded and nonempty when  $P$  is nonempty.

For each local state  $j \in I_B$ , write  $\lambda_j = \lambda_{B,j}$  and set

$$\gamma_{aj} = \min(\{\lambda_j \beta_a\} \cup \{z_{ej} - y_j v_e : C_e = a, (e, j) \in \mathcal{O}\} \cup \{-z_{ej} + y_j v_e : -C_e = a, (e, j) \in \mathcal{O}\}). \quad (67)$$

The observation lists are empty for the merged state. Each  $\gamma_{aj}$  is a concave piecewise-affine function of the original coordinates. Write

$$Q_j = \{\theta : M\theta \leq \gamma_j\}, \quad \bar{\theta}_i = x_{e_i} - v_{e_i}, \quad (68)$$

where  $e_i$  is chord  $i$ . With the original domains and bridge equations enforced, theorem 3.2 reduces hull membership to

$$Q_j \neq \emptyset \quad (j \in I_B), \quad \bar{\theta} \in \sum_{j \in I_B} Q_j \quad (69)$$

for each block. At zero weight, the signed coordinate bounds in (67) force  $\theta = 0$ , or reveal an inconsistent observation. Thus all subsequent statements include zero weights.

## 6.2 Finite local feasibility certificates

Let  $\mathcal{K}$  contain the primitive generators of the extreme rays of

$$K = \{q \in \mathbb{R}_+^{|\mathcal{A}|} : M^\top q = 0\}. \quad (70)$$

These are the positive circuits of the row configuration.

**Lemma 6.1** (Unit local certificates). *Every  $q \in \mathcal{K}$  is the indicator vector of a minimally dependent row set of size at most  $s + 1$ . Moreover,*

$$Q_j \neq \emptyset \iff q^\top \gamma_j \geq 0 \quad (q \in \mathcal{K}). \quad (71)$$

*A violated test yields an affine original-coordinate cut with unit product coefficients by selecting active branches in (67).*

*Proof.* For an extreme nonnegative dependence, the dependence space on its support must be one-dimensional. Otherwise a linearly independent dependence could perturb its strictly positive support in both directions while remaining nonnegative, contradicting extremality. No proper subset is dependent: such a dependence would provide the same perturbation unless it were a multiple of the original vector, which has full support. Hence the support has at most  $s + 1$  rows. A cofactor formula on a full-rank set of columns of this minimally dependent row matrix expresses its primitive dependence by nonzero minors. Total unimodularity makes every nonzero coefficient  $\pm 1$ , and positivity makes them all 1.

Farkas' lemma says  $M\theta \leq \gamma_j$  is feasible exactly when  $q^\top \gamma_j \geq 0$  for every  $q \in K$ . The pointed polyhedral cone  $K$  is generated by its extreme rays, proving (71). Enumerating row subsets of size at most  $s + 1$  and retaining the one-dimensional positive minimal dependencies therefore constructs the entire finite library.

An observation occurs only in the two opposite row normals  $C_e$  and  $-C_e$ . If a minimal positive dependence uses both these normals, it is exactly that opposite pair. Selecting the same observation twice then cancels it. Otherwise each observation occurs at most once. Selecting active affine minima gives an affine upper bound on  $q^\top \gamma_j$  that equals its negative value at the candidate and is nonnegative on the hull. Its product coefficients are thus  $0, \pm 1$ .  $\square$

### 6.3 Edge directions and a complete support library

For each set of  $s - 1$  independent rows of  $M$ , take one orientation of its primitive integer null vector and deduplicate. Call the resulting set  $\mathcal{D}$ . For  $s = 1$  set  $\mathcal{D} = \{1\}$ .

**Lemma 6.2** (Uniform edge directions). *Every direction  $d \in \mathcal{D}$  satisfies*

$$d \in \{0, \pm 1\}^s \setminus \{0\}, \quad Md \in \{0, \pm 1\}^{|\mathcal{A}|}. \quad (72)$$

*The set has at most  $(3^s - 1)/2$  elements and includes each coordinate direction. It contains every edge direction of every nonempty  $Q_j$ , including lower-dimensional states.*

*Proof.* The cofactor null vector of  $s - 1$  independent rows has entries  $0, \pm 1$  by total unimodularity, and at least one entry is nonzero. It is therefore already primitive. Its scalar product with another row is, up to sign, the full determinant obtained by adjoining that row, again  $0, \pm 1$ . This proves (72) and the cardinality bound. Using all but one coordinate row gives each coordinate direction.

At a relative interior point of a one-dimensional face, the tight defining rows have rank  $s - 1$ . To justify this even without interior, their common nullspace is exactly the direction space of the minimal face: all nontight inequalities remain slack under sufficiently small motions in that nullspace. Rows tight throughout  $Q_j$  are included among them. Choosing  $s - 1$  independent tight rows gives the asserted edge direction. Point states have no edges to check.  $\square$

Form the central arrangement in objective space with hyperplanes  $d^\top h = 0$ ,  $d \in \mathcal{D}$ . Let  $\mathcal{R}$  contain both primitive orientations of the null vector of every  $s - 1$  independent directions from  $\mathcal{D}$ ; for  $s = 1$  use  $\mathcal{R} = \{-1, 1\}$ . Extra directions in this list cause no difficulty. The coordinate hyperplanes occur, so every full-dimensional closed chamber is pointed. Every extreme ray of such a chamber is generated by a member of  $\mathcal{R}$ , since its tight hyperplanes have rank  $s - 1$ . Cofactors and Hadamard's determinant bound give

$$|h_i| \leq H_s \quad (h \in \mathcal{R}). \quad (73)$$

Indeed each cofactor has order  $s - 1$  and ternary entries; dividing by the greatest common divisor to make it primitive cannot increase it.

**Lemma 6.3** (Finite support characterization). *For any finite family of nonempty bounded polytopes whose edge directions belong to  $\mathcal{D}$ , membership in their sum is equivalent to*

$$h^\top \bar{\theta} \leq \sum_j \sigma_j(h) \quad (h \in \mathcal{R}), \quad \sigma_j(h) = \max_{\theta \in Q_j} h^\top \theta. \quad (74)$$

*This includes lower-dimensional summands and sums. The empty sum has support zero on every ray and is the singleton zero.*

*Proof.* In the interior of a chamber, an objective is perpendicular to no possible edge. Its maximizing face in each summand is therefore a single vertex: a positive-dimensional face of a polytope has an edge. Along a line segment inside the chamber, the maximizing vertex cannot change without a nonunique maximizer at an intermediate objective. Thus each support function is linear on the chamber, and by continuity on its closure. Its sum is linear there as well. Every vector in a pointed polyhedral chamber is a nonnegative combination of its extreme-ray generators. The inequalities on  $\mathcal{R}$  therefore imply the support inequality for every objective in every chamber closure, which covers all objective space. The support characterization of a nonempty compact convex set completes the proof. For the empty sum the same argument gives all inequalities  $h^\top \bar{\theta} \leq 0$ , forcing zero.  $\square$

This is the classical edge-direction/zonotope refinement argument [10, 15, 16]. Network-flow edge directions are also studied by Onn et al. [17]. The proof is included because uniformity over the state right-hand sides and lower-dimensional slices is essential here.

#### 6.4 Exact support formulas and bounded dual multipliers

For  $h \in \mathcal{R}$ , let  $\mathcal{B}(h)$  contain every nonsingular  $s$ -row submatrix  $B$  of  $M$  for which

$$q_B = B^{-\top} h \geq 0. \quad (75)$$

Rows and components have the same chosen order; extending  $q_B$  by zero gives a multiplier on all rows of  $M$ .

**Lemma 6.4** (Support bases and multiplier bound). *For every nonempty state domain,*

$$\sigma_j(h) = \min_{B \in \mathcal{B}(h)} q_B^\top \gamma_{B,j}. \quad (76)$$

*The list  $\mathcal{B}(h)$  is nonempty, and every retained multiplier is an integer satisfying*

$$0 \leq (q_B)_i \leq H_s. \quad (77)$$

*Proof.* The support LP has dual  $\min\{q^\top \gamma_j : M^\top q = h, q \geq 0\}$ . Signed coordinate rows make it feasible for every  $h$ . The primal is nonempty and bounded, so duality gives a finite attained optimum. There is an optimal solution with linearly independent supporting columns of  $M^\top$ . Indeed, a dependence on a positive support permits both signs of a small feasible perturbation; at an optimum its objective change must be zero. Moving until a component vanishes reduces the support, and repetition yields independence. Extend those rows to a basis of  $\mathbb{R}^s$ ; the added multipliers are zero. This proves (76), also for a degenerate or lower-dimensional primal.

Total unimodularity gives  $\det B = \pm 1$ , so  $q_B$  is integral. The stronger magnitude bound uses the defining directions of the ray rather than a triangle inequality on  $B^{-\top} h$ . Choose independent  $d_1, \dots, d_{s-1} \in \mathcal{D}$  whose primitive orthogonal vector is  $h$ . By (72), every  $Bd_i$  is ternary, and these vectors remain independent. The integer vector  $B^{-\top} h$  is orthogonal to them and primitive, because a unimodular transformation preserves primitive integer vectors. Their cofactor vector, divided by its common divisor, therefore equals  $B^{-\top} h$  up to sign. Hadamard's bound proves (77). Rank one is immediate.  $\square$

**Theorem 6.5** (Bounded-rank separation and coefficients). *For maximum cyclic block rank  $r$ , the sparse hull has a finite linear description whose flow and observed-product coefficients are integers of magnitude at most  $H_r$ . The sharper bound  $H_{r_B}$  applies to inequalities from block  $B$ . Simplex coefficients may contain rational network data; the bound is in the specified rational row scaling.*

*After observation grouping, exact membership and original-coordinate linear separation require*

$$2^{O(r^2)}(|V| + |E| + m + |\mathcal{O}|) \quad (78)$$

*rational arithmetic operations, including construction of finite block libraries. No online LP is required. Ungrouped observations may add  $O(|\mathcal{O}| \log |\mathcal{O}|)$  sorting comparisons.*

*Proof.* Use the original domains and bridge equations, the local circuit tests (71), and the support tests (74), evaluating each support by (76). Their equivalence to (69) proves completeness. Each support expression is a minimum of affine functions, because its multipliers are nonnegative and its endpoints are affine minima. If a support test fails, choose a minimizing basis and active endpoint branches separately for each state. The resulting affine majorant agrees with the support sum at the candidate, so it is a valid violated cut.

A nonsingular row basis cannot contain opposite normals. An observation therefore appears at most once in a state's selected support expression, with coefficient magnitude at most  $H_s$  by lemma 6.4. Product indices differ between states. The aggregate uses distinct original chord coordinates with coefficients  $h_i$ , bounded by (73). Local circuit cuts have unit product coefficients, and the original domains and bridge equations have unit flow/product coefficients. Expanding all branches gives the asserted finite description. Adding or clearing rational simplex denominators is not part of this normalization.

There are at most  $3^s - 1$  signed normals and  $(3^s - 1)/2$  edge directions. Enumerating their subsets of size at most  $s + 1$ , all candidate support rays, and all normal bases takes  $2^{O(s^2)}$  operations and storage. The product of the ray and basis counts has the same bound. All library integers and their encoding lengths depend only on  $s$ .

Fundamental cycles can be constructed in  $O(r|E|)$  graph/arithmetic operations. Repeated normals are combined in parameter-dependent work per arc, and each observation updates two endpoint entries. A block with  $a_B$  observed labels then needs  $2^{O(r_B^2)}(a_B + 1)$  arithmetic operations to initialize and check its local states and evaluate all supports. Since  $\sum_B a_B \leq |\mathcal{O}|$  and there are at most  $|E|$  blocks, summing these bounds proves (78).  $\square$

Online operations are rational affine evaluations, comparisons, and sums with parameter-bounded integer multipliers. Their encoding lengths are polynomial in the total input/candidate encoding length and the library encoding length. The arithmetic bound does not charge unbounded integers as single-bit objects. It also does not assert that a universal library is computationally attractive at moderate rank: the parallel-path oracle can be preferable even when its block ranks are unbounded.

## 6.5 Constructive recovery without an online LP

The finite supports can also provide a direct constructive decomposition. The following method deliberately uses a larger parameter factor than separation.

**Theorem 6.6** (Finite-basis compact recovery). *A rational hull point with maximum cyclic block rank  $r$  admits a compact rational decomposition into at most  $m + 1$  graph points, computable without an online LP in*

$$2^{O(r^3)}(|V| + |E| + m + |\mathcal{O}|) \quad (79)$$

arithmetic operations, including parameter-dependent preprocessing and with the same observation-grouping proviso as [theorem 6.5](#). The output and intermediate rational numbers have polynomial encoding length in the input/candidate and library encoding lengths. Dense state-flow output has the separate cost stated in [proposition 5.4](#).

*Proof.* Fix a block of rank  $s$ . Compute every state support on  $\mathcal{R}$  as above, order the local states, and precompute suffix sums  $S_i(h) = \sum_{j>i} \sigma_j(h)$ . At step  $i$ , let  $t$  be the remaining aggregate, which by induction belongs to  $\sum_{j \geq i} Q_j$ . The permissible next state vector is in the nonempty polytope

$$W_i(t) = \{\theta : M\theta \leq \gamma_i, \quad -h^\top \theta \leq S_i(h) - h^\top t \quad (h \in \mathcal{R})\}. \quad (80)$$

By [lemma 6.3](#), this is exactly  $Q_i \cap (t - \sum_{j>i} Q_j)$ . It is bounded because it is contained in  $Q_i$ . Even if lower-dimensional, a nonempty bounded polytope has a vertex, and its active defining normals at a vertex have full ambient rank: otherwise a nonzero null direction would allow small motions in both directions without violating any inequality.

The normal list  $N$  of (80) consists only of the rows of  $M$  and  $-\mathcal{R}$  and is fixed before the candidate is given. Enumerate all nonsingular  $s$ -row submatrices of  $N$  and precompute their inverses. For each basis, solve its equations using the current right-hand side and test the resulting vector against every row of (80). At least one candidate is a feasible vertex by the preceding paragraph. Choose the first, subtract it from  $t$ , and continue. The last suffix is the singleton zero, so the construction ends with zero remaining aggregate. This proves exact state recovery without solving an online LP.

The normal list has  $2^{O(s^2)}$  rows. Its  $s$ -row basis enumeration, including all feasibility checks per step, costs  $2^{O(s^3)}$  arithmetic operations and storage. The earlier support computation is cheaper. There are  $a_B + 1$  local states per block, so summing the same sparse state-count bound as in [theorem 6.5](#) proves (79).

For completeness, sequential division does not hide an exponential encoding cost in the number of states. The fixed normal entries are integers bounded by  $H_s$ . Every inverse basis is its integer adjugate divided by a nonzero integer determinant of parameter-bounded encoding length. All initial state endpoints and supports have a common rational denominator whose encoding length is polynomial in the input/candidate length and the library length. At each chosen step, a common denominator for the remaining aggregate need only be multiplied by that step's basis determinant. After  $a_B + 1$  steps its encoding length has therefore grown additively by at most  $a_B + 1$  parameter-bounded terms. Numerators also have polynomial length: state points are within their finite base bounds, remaining aggregates are sums of such points, and rejected basis candidates are obtained from these right-hand sides by fixed parameter-bounded matrices. This proves the claimed intermediate and output encoding bound.

Finally, normalize positive-weight explicit state coordinates and use the normalized merged state as the default for its constituent global labels. The zero-weight and unused-default conventions of [proposition 5.4](#) apply unchanged. Expand through  $C$  and use [theorem 3.2](#) to obtain graph points with the global simplex weights. A block coordinate has  $s$  entries, so defaults and observed-label exceptions give a compact representation within the stated bound.  $\square$

## 6.6 Coordinate sections and necessary coefficients

We next state a coefficient-transfer observation that will also apply to the more general network constructions below. A *coordinate plane* fixes all original coordinates except two designated coordinates  $U, V$ ; it does not rescale or project additional free coordinates.

**Lemma 6.7** (Local section forces an ambient coefficient ratio). *Let  $P$  be a polytope and  $L$  a coordinate plane with free coordinates  $U, V$ . Suppose that in an open neighborhood of a point  $(U_0, V_0)$ , the section  $P \cap L$  is exactly*

$$\alpha U + \beta V \leq \gamma, \quad \alpha U_0 + \beta V_0 = \gamma, \quad (\alpha, \beta) \neq (0, 0). \quad (81)$$

*Every finite linear inequality description of  $P$ , together with valid affine equations, contains an inequality whose coefficients on  $U, V$  are a positive multiple of  $(\alpha, \beta)$ . In particular, a relative facet description contains such an inequality. Adding any equation valid on  $P$  to a facet representative cannot change those two coefficients.*

*Proof.* The section has two-dimensional interior near  $(U_0, V_0)$ . Consequently every valid affine equation restricts to an identity on  $L$ : its coefficients on  $U, V$  are zero, and its fixed-coordinate constant is zero. Restrict a finite inequality description to  $L$ . At the boundary point, at least one nonconstant restricted inequality must be active. Otherwise all nonconstant inequalities are strictly satisfied in a neighborhood, and constant inequalities cannot produce the asserted boundary. An active valid inequality must vanish on the local boundary line, because feasible points occur in both tangent directions. Its normal is therefore proportional to  $(\alpha, \beta)$ , and feasibility on the interior side makes the multiple positive. The argument applies to a finite relative facet description as well. The previously proved zero coefficients of valid equations establish representative invariance.  $\square$

## 6.7 Sharp rank-three examples with sparse observations

Use vertices 0, 1, 2, 3 with every edge of  $K_4$  directed toward the larger vertex, order the arcs as (12, 13, 23, 01, 02, 03), and set every capacity to one. The tree is 01, 02, 03, with chord matrix

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & -1 \end{pmatrix}. \quad (82)$$

The following construction uses only two explicit simplex variables and five observed products. Unlike the source–sink example in [example 4.7](#), it has multiple supplies and demands.

Take reference flow and balances

$$v = (1/2, 1/2, 1/2, 1/4, 1/2, 1/2), \quad b = Av = (-5/4, -3/4, 1/2, 3/2)^\top. \quad (83)$$

Fix  $y_1 = y_2 = 1/3$ , so the residual weight is also  $1/3$ , and fix

$$\bar{\theta} = (1/6, 1/24, 1/8), \quad \bar{x} = (2/3, 13/24, 5/8, 11/24, 11/24, 1/3). \quad (84)$$

Observe exactly

$$\mathcal{O} = \{(12, 1), (13, 1), (02, 1), (23, 2), (01, 2)\}. \quad (85)$$

Leave  $U = z_{12,1}$  and  $V = z_{13,1}$  free, and fix the other observations to  $z_{02,1} = z_{23,2} = 1/6$  and  $z_{01,2} = 1/12$ . Put  $p = U - 1/6$ ,  $q = V - 1/6$ . The observations give

$$\theta^1 = (p, q, p), \quad \theta^2 = s(1, -1, 0). \quad (86)$$

All state deviations on arcs other than 01 lie in  $[-1/6, 1/6]$ ; the interval on 01 is  $[-1/12, 1/4]$ . In particular, the residual state obeys  $\theta_1^0 \leq 1/6$  and  $\theta_2^0 + \theta_3^0 \leq 1/6$ . For  $h = (1, 1, 1)$ , these bounds give  $h^\top \theta^0 \leq 1/3$ , while  $h^\top \theta^2 = 0$  and  $h^\top \bar{\theta} = 1/3$ . Every feasible point in the section therefore satisfies

$$2p + q \geq 0, \quad 2U + V \geq 1/2. \quad (87)$$

Conversely, suppose  $|p|, |q| < 1/96$  and  $w = 2p + q \geq 0$ . Choose

$$s = q/2, \quad \theta^0 = (1/6 - p - q/2, 1/24 - q/2, 1/8 - p). \quad (88)$$

These vectors sum to (84). The first state's arc deviations are  $(p, q, p, p + q, 0, -p - q)$ , all within its scaled bounds. The second state's arc deviations have magnitudes at most  $|q|/2$  and are zero on 01. The residual arc deviations are

$$C\theta^0 = (1/6 - w/2, 1/24 - q/2, 1/8 - p, 5/24 - p - q, -1/24 + q/2, -1/6 + w/2). \quad (89)$$

Here  $0 \leq w < 1/32$ . The first and last entries lie in their closed intervals  $[-1/6, 1/6]$ . The second, third, and fifth entries lie strictly inside those intervals. The fourth lies in  $(3/16, 11/48)$ , contained in  $[-1/12, 1/4]$ . Thus all scaled bounds hold, as do the balances and observations. Disaggregation proves that the section is exactly the half-plane (87) in the asserted neighborhood.

**Corollary 6.8** (Rank-three sharpness with two simplex variables). *The bound  $H_3 = 2$  in [theorem 6.5](#) cannot be replaced by one, even for a unit-capacity acyclic  $K_4$  with two explicit simplex variables and five observed products. Every finite original-coordinate description of this hull contains an inequality whose coefficients on  $U, V$  have ratio two.*

*Proof.* Apply [lemma 6.7](#) with the local normal  $(-2, -1)$  at  $(U, V) = (1/6, 1/6)$ . An integer flow/product coefficient description with all magnitudes at most one cannot contain this ratio, whereas [theorem 6.5](#) supplies a description with bound two.  $\square$

*Remark 6.9* (A symmetric-reference variant). There is also a seven-observation construction with the same graph and  $C$ , the uniform reference  $v = (1/2, \dots, 1/2)$ , and balances  $(-3/2, -1/2, 1/2, 3/2)^\top$ . Use three explicit variables  $y_1 = y_2 = y_3 = 1/4$ , aggregate cycle vector  $(1/8, 0, 1/8)$ , and therefore aggregate flows  $(5/8, 1/2, 5/8, 5/8, 1/2, 3/8)$ . Observe

$$(12, 1), (13, 1), (02, 1), (12, 2), (03, 2), (23, 3), (01, 3),$$

fix all but  $U = z_{12,1}, V = z_{13,1}$  to  $1/8$ , and put  $p = U - 1/8, q = V - 1/8$ . The explicit states are  $(p, q, p), t(0, 1, -1)$ , and  $s(1, -1, 0)$ ; every state has  $|C\theta^j| \leq 1/8$ . The residual support in direction  $(1, 1, 1)$  is at most  $1/4$ , so  $2p + q \geq 0$ , or  $2U + V \geq 3/8$ , is necessary. For  $|p|, |q| < 1/32$  and  $w = 2p + q \geq 0$ , the choices

$$t = -q/2, \quad s = q/2, \quad \theta^0 = (1/8 - w/2, 0, 1/8 - w/2)$$

prove sufficiency locally. The first state's arc deviations are  $(p, q, p, p + q, 0, -p - q)$ , the two line-state magnitudes are less than  $1/64$ , and the residual arc deviations are  $(a, 0, a, a, 0, -a)$  with  $5/64 < a = 1/8 - w/2 \leq 1/8$ . This variant has completely symmetric state bounds. Changing the reference on 01 in the five-product construction makes the residual support face an edge instead of a point, allowing one fewer explicit line state.

This necessity result concerns actual retained product coordinates and is invariant under adding the hull's affine equations. It is stronger than finding a dual basis with multiplier two. The related Khademnia and Davarnia [11, Example 2] exhibits a projection-cone aggregation weight two on a different graph with balance inequalities; it is an antecedent for weighted cancellation, but does not by itself give this equality-network facet-ratio conclusion.

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